



LAB PRACTICE № 4: CONSTRUCTOR, INHERITANCE REVIEW

COMP1020: OOP, Algorithms, and Data Structures

Week 04

Lab Practice Submission Instructions:

- This is an individual lab practice and will typically be assigned in the computer laboratory.
- Your program should work correctly on all inputs. If there are any specifications about how the program should be written (or how the output should appear), those specifications should be followed.
- Your classes and methods should be appropriately commented. However, try to avoid making your code overly busy (e.g., include a comment on every line).
- Variables, methods and classes should have meaningful names, and should be followed Java programming convention.
- Academic honesty is required in all work you submit to be graded. Although you can discuss among yourself at high level how to solve the lab problem, you should **NOT** copy or share your code with other students to avoid plagiarism issues.
- Use the template provided to prepare your solutions.
- You should upload your .java file(s) to both the Canvas and CMS Autograder **before the end of the laboratory session** unless the instructor gave a specified deadline.
 - (a) No Canvas submission will get 0 point (of course).
 - (b) **No CMS Autograder submission will get 20% deduction.**
- Submit separate *.java file for each Lab problem with the exact file name specified in each lab question (do not add your student ID or your name). For example: **HelloWorld.java**.
- Late submission of lab practice without an approved extension will incur the following penalties:
 - (a) No submission by the deadline will incur 0.25 point deduction for each problem (most of the problems are due at the end of the lab session).
 - (b) The instructor will deduct an additional 0.25 point per problem for each day past the deadline.
 - (c) The penalty will be deducted until the maximum possible score for the lab practice reaches zero (0%) unless otherwise specified by the instructor.

Learning Outcome

Topics: Polymorphism, Abstract classes and methods. Interfaces. Overriding and casting in Java.

Outcome: Acquire the skill of implementing and using polymorphic programming structures, abstract classes and methods, interfaces, overriding, and casting. Be able to define and use collections and generics in Java.

CMS Autograder

Direct access to this CMS lab is followed: <https://www.cmsvinuni.online>

Note: Due to the current limitation of the Autograder, you must comment out all the `package` declaration before submission.

Problem 01: Object moving in a 2D-space

[CMS Autograder is required] Write a program and name it `MovingObject.java`: An object moving in a 2D-space which. The location of the object is represented by 2 integers (x,y) which are x-coordinate and y-coordinate. Each step, the object can perform 1 of the 4 actions below:

- `MOVE_LEFT`: move left, x is reduced by 1
- `MOVE_RIGHT`: move right, x is increased by 1
- `MOVE_UP`: move up, y is increased by 1
- `MOVE_DOWN`: move down, y is decreased by 1

Perform the moving simulation of the object (a sequence of moving action above) and print the final location of the object.

Design a class representing the object with

- Two properties: x and y which are integers
- Five methods: `moveLeft()`, `moveRight()`, `moveUp()`, `moveDown()`, `printLocation()`

Input

- Line 1: contains 2 integers x and y ($0 \leq x, y \leq 10000$) which are the initial location of the object.
- Each line (in subsequent lines) contains a string representing one of the 4 moving actions described above.
- The input is terminated by a line containing `#`

Output

- Write the location (x and y separated by a SPACE character)

Example See Figure 1

stdin	stdout
6 4 MOVE_DOWN MOVE_UP MOVE_LEFT MOVE_DOWN MOVE_LEFT #	4 3

Figure 1: Example testcase of Problem 01

Problem 02 - Cube Volume

[CMS Autograder is required]

1. Create a java file named `CubeVolume.java` which consists of two classes: a default class `RectangularPrism` and a public class `CubeVolume`, where class `CubeVolume` inherits class `RectangularPrism`.
2. In the parent class, create 3 instance variables with datatype `double`, `length`, `width`, `height`, which are the length, width and height of a rectangular prism. Also, define method `getSurfaceArea()` which returns the surface area of a rectangular prism and method `getVolume()` which returns the volume of a rectangular prism.
3. In the child class, create an instance named `color`, which is the color of a cube; a method `printColor()` to print the color of a cube and method `cubeInformation()` to show a cube's information.
4. In the main method, complete your code to allow the user to perform the following operations:
 - create a cube with length `1.00` and color `blue` if the user enter a **space key**.
 - create a cube with length `s` and color `blue` if the user enter **a number (any type)**.
 - create a cube with length `s` and color `c` if the user enter **a number (any type) and a color**.
5. Next, create an instance of class `CubeVolume` using information provided by the user.

```
Case 1. Enter a space key to create a cube with length 1.00 and color blue, or
Case 2. Enter one number to create a cube with length s and color blue, or
Case 3. Enter a number and a color to create a cube with length s and color c:
2.5 Red
Cube's detail:
length: 2.50
width: 2.50
height: 2.50
Computing the surface area...
The surface area of the cube is 37.50
Computing the volume...
The volume of the cube is 15.63
The cube is Red
```

Figure 2: Problem 02 - Example a

```
Case 1. Enter a space key to create a cube with length 1.00 and color blue, or
Case 2. Enter one number to create a cube with length s and color blue, or
Case 3. Enter a number and a color to create a cube with length s and color c:
1.5
Cube's detail:
length: 1.50
width: 1.50
height: 1.50
Computing the surface area...
The surface area of the cube is 13.50
Computing the volume...
The volume of the cube is 3.38
The cube is blue
```

Figure 3: Problem 02 - Example b

```
Case 1. Enter a space key to create a cube with length 1.00 and color blue, or
Case 2. Enter one number to create a cube with length s and color blue, or
Case 3. Enter a number and a color to create a cube with length s and color c:
█
Cube's detail:
length: 1.00
width: 1.00
height: 1.00
Computing the surface area...
The surface area of the cube is 6.00
Computing the volume...
The volume of the cube is 1.00
The cube is blue
```

Figure 4: Problem 02 - Example c